A Latent Oscillator Measurement Model to Simulate Emotional-Expression Score Dynamics in Video

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Abstract

Facial-expression classifiers convert video into multivariate time series of scores with measurement error from classifiers, videos, and recording conditions. Empirical score series cannot establish whether the score channels reflect a smaller set of latent expressive processes or whether an analysis would recover those processes. We introduce the Latent Oscillator Measurement Model (LOMM), a data-generating model that separates latent dynamics, time-varying activity, and a factor-analytic observation model. The latent processes are damped, undamped, or amplifying linear oscillators. LOMM generates bounded scores or continuous indicators. Study 1 used four-fold cross-fitting with scores from 100 MAFW videos to calibrate LOMM and evaluate generated series on held-out videos. Median plausibility and coverage were 0.970 and 0.920 for LOMM, versus 0.510 and 0.370 for a calibrated static logistic-normal generator. Study 2 tested whether Dynamic Exploratory Graph Analysis (DynEGA), static EGA, GraphicalVAR, and GIMME recovered a known dimensional structure from continuous indicators generated by LOMM. At 100 observations per clip, with failed or timed-out fits counted as incorrect, correct-dimension recovery was 0.939 for DynEGA, 0.884 for static EGA, 0.777 for GraphicalVAR, and 0.176 for GIMME. Replacing the common fixed initialization with independent stationary starts for stable dimensions and bounded independent starts for amplifying dimensions reduced recovery for DynEGA, static EGA, and GraphicalVAR. The primary initialization was optimistic relative to this alternative. LOMM provides a controlled test of whether an analysis recovers a specified latent structure before score patterns are interpreted psychologically.

Introduction

Studying how emotional expression unfolds over time requires measurements that capture changes in visible behavior across successive moments. Automated emotion detection makes this task tractable by converting video into multivariate time series of facial-expression scores: each frame yields a score for every emotion label in a chosen set, allowing researchers to examine expressive variation at a temporal resolution that manual coding cannot match. Earlier classifiers restricted this analysis to categories represented in their training data, but vision-language models such as CLIP can score labels supplied as text (Foteinopoulou & Patras, 2023; Li et al., 2023; Radford et al., 2021). Interpreting the resulting series requires accounting for how those scores are produced. They depend on the model, label set, video-processing pipeline, image quality, and recording conditions, and their accuracy varies with model adaptation, label choice, and evaluation data (Foteinopoulou & Patras, 2023; Li et al., 2023). In a small demonstration on FindingEmo images, zero-shot CLIP achieved 41.7% accuracy and a macro-F1 of 0.331 (Tomašević et al., 2026). On the Image-Emotion dataset, fine-tuned Emotion-CLIP reached 67% accuracy and a macro-F1 of 0.64 in a reported 10-fold evaluation (Bondielli & Passaro, 2021). Classifier scores therefore provide fallible measurements of visible expression whose patterns can reflect expressive behavior, the measurement process, or both. Their use in emotion research requires a clear distinction between the visible expression being scored and the subjective emotional experience that those scores alone cannot establish.

This raises a psychometric question. Score channels can be analyzed as separate variables, or they can be treated as fallible indicators of a smaller number of latent expressive processes (Mehu & Scherer, 2015; Nesselroade et al., 2002; Russell, 2003). The number of classifier labels does not establish how many distinct processes the channels represent. Treating overlapping channels as separate dimensions can overstate that number, while combining indicators of distinct processes can conceal differences. Dimensionality assessment therefore concerns both the number of latent processes and the assignment of channels to those processes.

The observable associations among indicators can change even when the generating dimensions and indicator assignments remain fixed. Dynamics, activity, measurement error, and observation length affect how clearly the dimensional structure appears in a recorded series. Indicators of the same process may share little observable variation when that process is weakly expressed or measurement error is large, and measurement error can change the structure recovered by dynamic methods (Kankaanpää et al., 2026). Changes in activity can also produce changes in the indicators without a discontinuity in the underlying latent level. Evaluating recovery therefore requires conditions in which these sources of variation can be controlled while the generating structure remains known.

We introduce the Latent Oscillator Measurement Model (LOMM) for this purpose. LOMM combines the damped linear oscillator model of intraindividual dynamics (Boker, 2015; Boker & Nesselroade, 2002) with an activity layer and a factor-analytic observation model (Nesselroade et al., 2002; Zhang et al., 2008). It simulates latent processes, specifies how strongly each process is expressed at each observation, and maps the expressed processes onto observed indicators with measurement error. These components and the observation length can be varied while the generating dimensions and indicator assignments remain fixed, allowing researchers to examine when a specified structure is recovered and which conditions obscure it. Empirical score series can guide dimensionality assessment and the choice of simulation settings. Direct evaluation of recovery requires simulated data for which the generating structure is known.

We use LOMM in two studies (Figure 1). In Study 1, we extract emotion-score series from videos in the MAFW dataset, a multimodal database for dynamic facial-expression recognition in the wild (Liu et al., 2022), and evaluate how closely LOMM reproduces their statistical properties. Each of four folds uses 75 MAFW videos to calibrate LOMM and 25 videos to evaluate the generated series. A static logistic-normal generator provides a comparison without temporal dependence. Eleven statistics computed on scores and log-ratio coordinates define series-level plausibility and coverage.

Study 2 evaluates recovery of a known dimensional structure from LOMM-generated data. LOMM generates continuous indicators with a known number of dimensions and a known assignment of indicators to dimensions, and four network methods are used to recover this dimensional structure: Dynamic Exploratory Graph Analysis (DynEGA), static EGA, GraphicalVAR, and GIMME. Each method estimates a network among the indicators, which is then partitioned into communities. The question is whether the recovered communities match the structure that generated the data. Static EGA ignores temporal order and is computationally inexpensive; it shows how much structure is recoverable from contemporaneous indicator levels alone. Study 2 also tests how dimensional recovery depends on initialization and the transformation of continuous indicators into bounded scores. One analysis compares the common fixed initialization with independent stationary starts for stable dimensions and bounded independent starts for amplifying dimensions. A second tests whether the indicator grouping is preserved after the indicators are transformed to bounded scores that sum to one, as classifier scores do.

The Latent Oscillator Measurement Model

Emotion labels can share broader psychological functions, including engagement with desirable opportunities, defense against threats, and attention to novel events (Bradley, 2009; Lang & Bradley, 2010). These

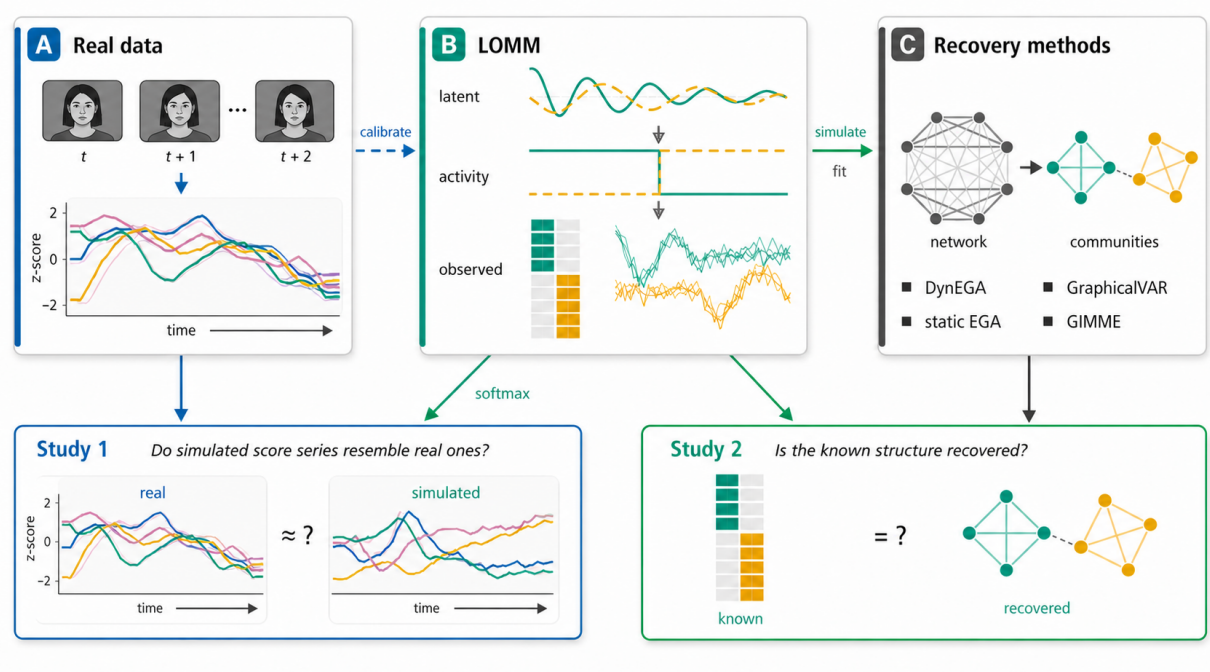


Figure 1: Overview of the two studies. (A) Real data: a classifier scores each sampled frame of an MAFW video, giving one score series per label. (B) LOMM: latent oscillators, an activity schedule that sets how strongly each is expressed at each observation, and observed indicators with a known assignment to dimensions (colored blocks). (C) Recovery methods: DynEGA, static EGA, GraphicalVAR, and GIMME each estimate a network among the indicators and group them into communities. Study 1 uses MAFW calibration videos to set the generator and compares simulated score series with MAFW evaluation series. Study 2 fits the four methods to simulated indicators and compares the recovered communities with the known assignment.

functions motivate an illustrative LOMM simulation with three latent dimensions labeled approach-affiliation, threat-defense, and orienting-novelty. Consider the threat-defense dimension: fear and anxiety both concern responses to threat, although they involve different defensive functions (McNaughton & Corr, 2004). A researcher could therefore specify fear, anxiety, worry, and apprehension score channels as indicators of a shared threat-related process. In the simulation, these channels covary as that process changes over time, while each also contains measurement error. Other channels measure the remaining two dimensions. The recovery task is to identify the three dimensions and their indicator assignments. This grouping is a theory-informed simulation hypothesis; whether empirical classifier scores exhibit it requires separate investigation.

LOMM generates data in three layers (Figure 2). The latent layer describes how each latent dimension changes over time. The activity layer determines how strongly that dimension is expressed at each observation. The observation layer creates multiple indicators with prespecified loadings and measurement error. Keeping these layers separate allows a researcher to change the dynamics, the expression pattern, or the measurement quality while holding the other components fixed.

Notation. Suppose we have N video clips, each represented by T sampled frames. Scoring each sampled frame produces one multivariate observation. We index clips by $i = 1, \ldots, N$ and observations within a clip by $n = 1, \ldots, T$. Observations occur at times $t_n = (n-1)\Delta$, where Δ is the interval between successive observations. Latent dimensions are indexed by $f = 1, \ldots, m$. Continuous indicators are indexed by $j = 1, \ldots, p$; in Study 2, $p = mk$ with k indicators per dimension. Bounded scores have J channels. Bold symbols are vectors or matrices.

For clip i and latent dimension f, let $x_{if}(t)$ denote the latent level and $\dot{x}_{if}(t)$ its rate of change. LOMM treats each dimension as a linear second-order oscillator:

$$\ddot{x}_{if}(t) = \eta_f x_{if}(t) + \zeta_f \dot{x}_{if}(t) + q_{if,n}, \qquad t \in [t_n, t_{n+1}). \tag{1}$$

Equation 1 is the damped linear oscillator used in research on intraindividual dynamics (Boker & Nesselroade, 2002). Affective-dynamics models describe how emotional states depart from and return toward a characteristic baseline. Damped oscillator models represent this process through a tendency to return toward baseline and a separate tendency for ongoing fluctuations to diminish (Chow et al., 2005). LOMM uses these dynamics to generate latent expressive trajectories with different patterns of persistence, recovery, and amplification.

The position-feedback parameter η_f determines how strongly a deviation from baseline changes the direction of the trajectory. When $\eta_f < 0$, values above baseline produce a downward acceleration and values below baseline

produce an upward acceleration. The velocity-feedback parameter ζ_f determines whether ongoing changes are slowed or reinforced. Negative values oppose the current direction of change and damp fluctuations; zero adds no damping; positive values reinforce the change and amplify it. These parameters describe the simulated dynamics; interpreting them as individual differences in emotion regulation would require validation against independent psychological measures. The term $q_{if,n}$ is a random acceleration input, drawn independently for each interval between observations n and $n+1$ and held constant within that interval; it produces irregular peaks and reversals in the trajectory.

When both feedback parameters are negative, the process tends to settle toward its baseline after a disturbance. It may approach the baseline smoothly or move above and below it with progressively smaller swings. New random inputs can keep these fluctuations going. Positive velocity feedback instead reinforces ongoing changes and can produce increasingly large swings. We include these growing trajectories to test whether the methods can recover the specified dimensions when signal amplitudes increase over a clip. Appendix C in the supplementary material reports how large the simulated amplitudes become. These conditions are evaluated over a fixed observation period; continued growth would eventually become implausible for sustained expressive behavior. Appendix A in the supplementary material gives the equations used to calculate each successive latent state, explains which parameter settings produce stable behavior, and describes how the starting values are chosen.

The activity layer represents how strongly a latent process contributes to the observed expression at a particular moment. Separating a process from its expression is motivated by evidence that outward emotional behavior can change without a corresponding change in reported experience, as in expressive suppression (Gross & Levenson, 1993). In LOMM, the nonnegative activity value $a_{if,n}$ scales the latent signal: $\tilde{x}_{if,n} = a_{if,n} x_{if,n}$. Small values weaken its contribution to the indicators, and larger values strengthen it. The activity schedule is specified by the researcher and can represent periods of weaker or stronger expression. It does not identify the psychological mechanism producing those changes. This layer matters for recovery because an abrupt change in activity can produce a large change in the observed indicators even when the latent level remains continuous.

The observation layer is a standard factor model:

$$\mathbf{z}_{in} = \mathbf{\Lambda}_i^{(z)} \tilde{\mathbf{x}}_{in} + \boldsymbol{\varepsilon}_{in}^{(z)}, \qquad \boldsymbol{\varepsilon}_{in}^{(z)} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Psi}_i). \tag{2}$$

Row j of $\mathbf{\Lambda}_i^{(z)}$ contains the loadings of indicator j on the m latent dimensions, and $\boldsymbol{\varepsilon}_{in}^{(z)}$ is measurement error with covariance $\mathbf{\Psi}_i$. Appendix A in the supplementary material gives the complete observation equations,

including the separate notation for continuous indicators and log-ratio coordinates and the optional clip-specific baseline and scale terms.

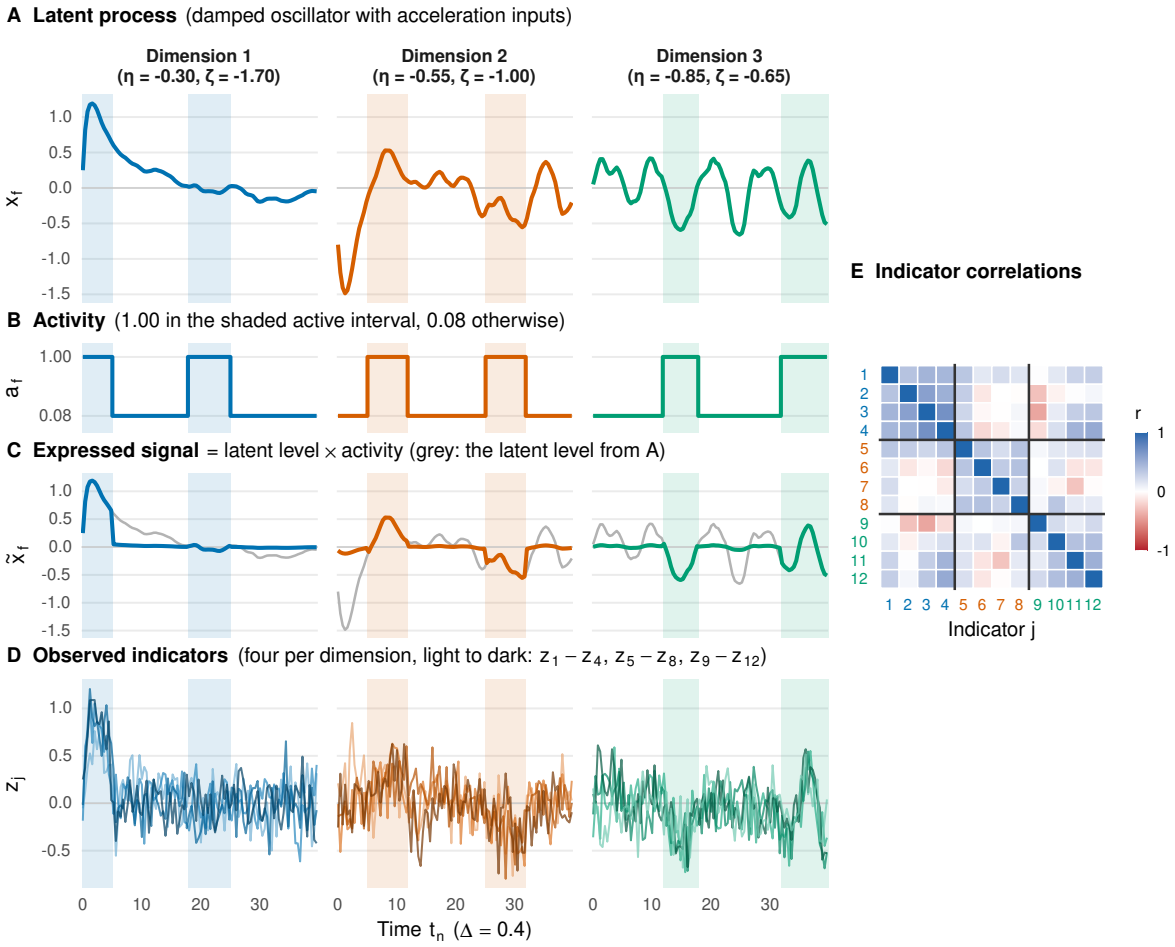


Figure 2: One simulated LOMM clip with three latent dimensions, $T = 100$ observations, and $\Delta = 0.4$. Each column follows one dimension from the latent layer to the observed indicators. (A) Latent trajectory; the oscillator parameters are given in the column headers, and each dimension has its own initial state and acceleration inputs. (B) Activity, 1.00 in the shaded active interval and 0.08 otherwise. (C) Expressed signal, the product of the latent level and the activity value, drawn over the latent level in grey. (D) The four observed indicators of each dimension in shades from light to dark, with primary loadings 0.65 to 0.85, secondary loadings of ±0.30 or ±0.35 on another dimension for nine of the twelve indicators, and residual SD 0.20. (E) Correlations among the twelve indicators from this clip, ordered by dimension.

Figure 2 shows the three layers for one simulated clip. In panel A the dimensions differ in their feedback parameters and start from different states: dimension 1 rises to an early peak and decays smoothly, dimension 2 starts in a trough and rebounds, and dimension 3 oscillates faster and is driven mainly by its acceleration inputs. Panel B shows the activity schedule, in which the dimensions take turns being active. Panel C shows the product of the two layers. A latent peak that falls inside an active interval is expressed at full size; the same peak outside the interval is reduced to 8% of its size and is barely visible in the indicators. The steps at

the boundaries of the active intervals appear in the indicators although the latent level remains continuous. Panel D shows the twelve indicators; each loads mainly on its own dimension, and nine also carry a smaller loading on another dimension. Panel E shows the correlation matrix among the twelve indicators from this clip. The three diagonal blocks reflect the indicator assignments, which is the structure that Study 2 asks the methods to recover. The weaker correlations between blocks can reflect both secondary loadings and sample covariation among the expressed latent trajectories. The secondary loadings in this example are larger than in the Study 2 cross-loading condition, which limits them to ± 0.10.

LOMM can output two types of observed variables. The first is the vector of continuous indicators $\mathbf{z}_{in}$, used in the primary Study 2 designs; there, the factor structure that generated the data is defined directly on the analyzed variables. The second is a vector of J positive scores that sum to one, which matches the form of classifier output and is used in Study 1. To produce bounded scores, LOMM maps the latent dimensions to a $(J-1)$-dimensional vector of log-ratio coordinates and applies the softmax function. Because scores that sum to one are compositional data, they are analyzed both as scores and in log-ratio coordinates, which remove the sum constraint. A separate Study 2 condition applies the softmax directly to the p continuous indicators and tests whether their original grouping is preserved. Appendix A in the supplementary material gives the softmax, log-ratio, and identification details.

In Study 1, MAFW calibration videos determine the LOMM observation-model geometry and generator settings. The primary Study 2 design sets between-clip baselines to zero and scales to one but lets loadings and residual variances vary slightly across clips. Estimating oscillator, activity, and measurement parameters from empirical data would require the sign, scale, temperature, and frequency constraints stated in Appendix A in the supplementary material.

Study 1: Comparing Simulated and Real Score Series

Study 1 examined whether LOMM, after calibration to MAFW, could generate classifier-score series with statistical properties observed in MAFW videos excluded from calibration. The analysis used 100 videos from MAFW, a large multimodal database of facial-expression video recorded in naturalistic settings (Liu et al., 2022). Frames were scored with the transforEmotion R package (Tomašević et al., 2026) using OpenAI's CLIP ViT-Large/14 model (setting `oai-large`), with up to 50 frames sampled per video. Eleven labels from the MAFW label set served as text prompts: anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment. For each scored frame, the eleven classifier-score channels sum to one.

Figure 3 shows the classifier-score series for one MAFW video. Frames without a detected face have missing scores, so each video was divided into segments of consecutive scored frames. The analysis contains 3,557 observations in 188 segments. Videos contribute 16 to 50 scored observations, with a median of 35; segment lengths range from 1 to 50 observations, with a median of 13.5. Appendix B in the supplementary material gives the face-detection, score-validity, missing-data, and sampling details.

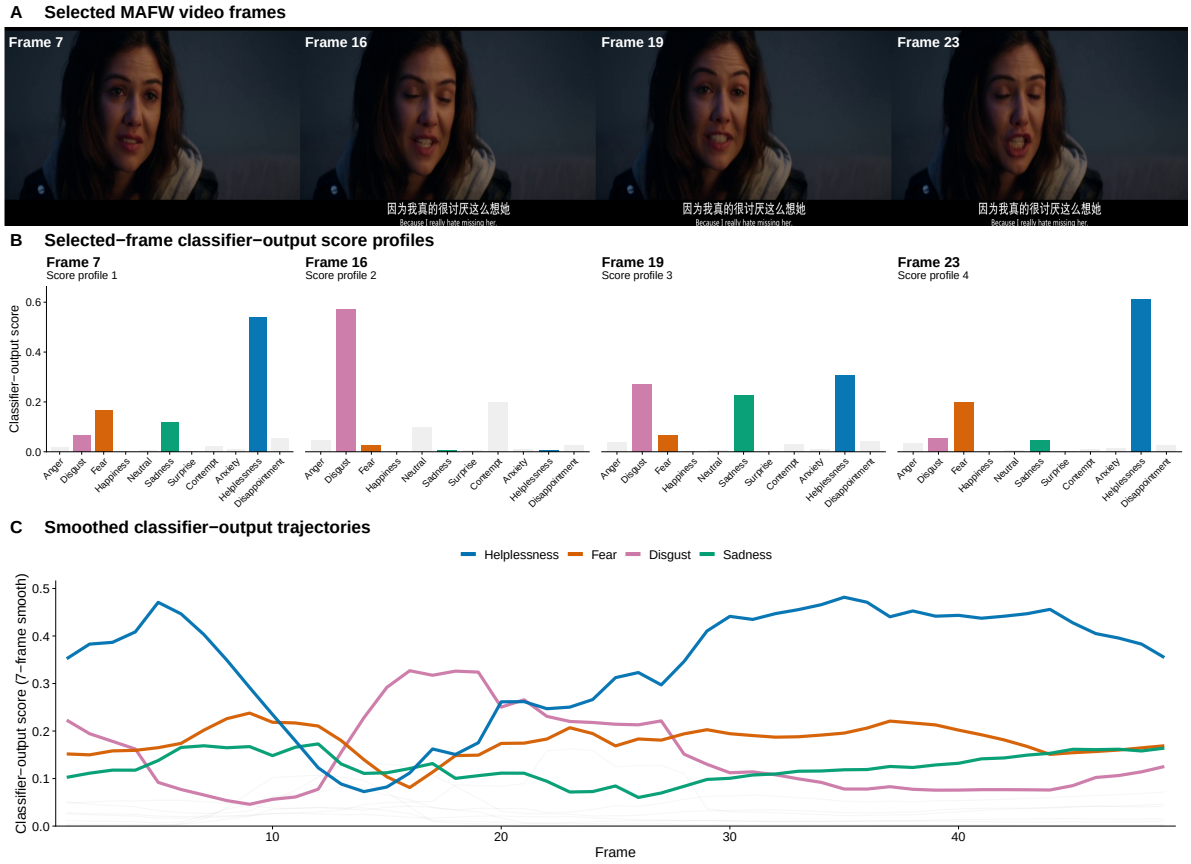


Figure 3: Classifier-score series from one MAFW video. Panel A shows four frames from the video, reproduced under the MAFW research-use terms. Panel B shows the eleven classifier scores for those frames. Panel C shows smoothed trajectories for the channels with the highest mean scores. Scores were generated with transforEmotion using OpenAI CLIP ViT-Large/14.

We represented each MAFW and simulated series with a broad panel of statistics covering location, dispersion, short-term change, temporal persistence, temporal irregularity, and cross-channel dimensionality. Eight statistics were computed within channel and video: median level; interquartile range of levels; interquartile range of observation-to-observation changes; lag-1 autocorrelation; and run-length, sample, permutation, and spectral entropy. Their channel-specific values were summarized by the median within each video. Effective rank, robust level variance, and robust change variance described all channels jointly. Each statistic was evaluated on the classifier scores and a log-ratio representation, producing 22 features. Channel statistics

used centered log-ratios (CLR), and the three joint statistics used isometric log-ratio (ILR) coordinates. The main analysis combines the complete feature panel in the plausibility and coverage measures. Appendix C in the supplementary material reports the metric-specific comparisons.

We used four-fold cross-fitting. In each fold, 75 MAFW videos formed the calibration set and 25 formed the evaluation set. The calibration set determined the LOMM measurement geometry and the dynamic and measurement parameters. LOMM then generated 100 datasets with the same observation positions and uninterrupted segment boundaries as the evaluation videos. We compared LOMM with a static logistic-normal generator calibrated from the same 75 MAFW videos. The static generator retained the estimated compositional structure, between-video heterogeneity, and observation pattern while drawing observations independently over time. This comparison estimates the gain associated with temporal generation beyond static calibration.

For each fold, the MAFW reference comprised 1,000 comparisons between two disjoint, category-stratified groups of 25 videos drawn from the 75 calibration videos. The 22 features were standardized using medians and robust scales estimated from the calibration set. Series similarity was measured by the root-mean-square Euclidean distance across these standardized features. The similarity threshold was the 95th percentile of the directional nearest-neighbor distances in the MAFW reference comparisons. Plausibility is the proportion of simulated series whose nearest MAFW evaluation series fell within this threshold. Coverage is the proportion of MAFW evaluation series whose nearest simulation fell within the threshold. Reported ranges are the 2.5th and 97.5th percentiles across simulation or reference replicates. Appendix B in the supplementary material gives the calibration, parameter selection, metric, and reference-sampling procedures.

Study 2: Recovery of Dimensionality and Indicator Assignments

Data-generating conditions. Study 2 asks whether four network-based methods recover the number of latent dimensions and their indicator assignments from LOMM-generated data. The primary comparison applies all four methods to the same simulated datasets, including a shorter-record condition. A larger factorial simulation evaluates DynEGA across additional combinations of dimensionality, indicators per dimension, and observation schedules. These analyses use continuous indicators. Each dataset contains N clips with T observations each and $p = mk$ indicators. All clips share the same indicator assignments and nominal measurement parameters, while allowing small differences in measurement strength across clips. In these comparison and factorial designs, each primary loading receives an independent $\mathrm{Unif}(-0.15, 0.15)$ deviation, and each residual variance receives an independent $\mathrm{Unif}(-0.01, 0.01)$ deviation.

A latent dimension can be present in the generating model yet contribute little observable variation during a short recording. Weak expression can leave its indicators dominated by measurement error. When dimensions alternate in prominence, each contributes strongly during only part of the recording, and transitions can change the associations among indicators. Increasing amplitudes can make later observations contribute disproportionately to those associations. The six scenario families examine recovery across these different patterns of expression while keeping the indicator assignments fixed (Table 1). Families A and B represent weak and strong expression with one dimension remaining dominant. Families C and D allow the dominant dimension to change, with more rapid switching in D. Families E and F introduce growing amplitudes, with either one dominant dimension or switching dominance. These families combine activity gains, switching schedules, and dynamic settings to represent different recovery challenges. Appendix B in the supplementary material gives the activity schedules and implementation details; Table B2 gives the numerical oscillator parameters.

Table 1: The six Study 2 scenario families. Numeric activity gains and dwell fractions are in Appendix B in the supplementary material; oscillator parameters are in Table B2.

Code	Family
A	Weak expression, one dominant dimension
B	Strong expression, one dominant dimension
C	Strong expression, switching dominant dimension
D	Strong expression, rapid switching
E	Amplifying dynamics, one dominant dimension
F	Amplifying dynamics, switching dominant dimension

Recovery procedures. Network methods provide a way to estimate both the number of dimensions and the assignment of indicators to them. In LOMM, indicators assigned to the same latent process share a changing signal. Under suitable measurement conditions, this shared variation can produce groups of strongly connected indicators in an estimated network. Community detection can therefore identify candidate dimensions without specifying their number in advance (H. F. Golino & Epskamp, 2017). Study 2 examines whether this correspondence survives changes in dynamics, expression strength, measurement error, and observation length.

The four procedures construct networks from different aspects of the same data: indicator levels, estimated changes, or residual associations after temporal modeling (Table 2). Every method receives the same simulated dataset. Each estimated structure is converted to an undirected network by a prespecified rule,

and a common community-detection step identifies groups of indicators. Comparing these groups with the generating assignments shows how well each representation preserves the indicator partition, while retaining each method's statistical model and data handling.

Table 2: The four Study 2 methods, the structure each supplies for the common grouping step, and its purpose in the recovery comparison.

Method	Structure used for grouping	Purpose in the recovery comparison
DynEGA	Partial-correlation network of estimated changes	Assess recovery from coordinated changes in the indicators
Static EGA	Partial-correlation network of contemporaneous levels	Assess recovery from associations among indicator levels
GraphicalVAR	Contemporaneous residual partial-correlation network	Assess recovery from residual associations after modeling lagged relations
GIMME	Average residual-covariance network across clip-level fits	Assess recovery from residual associations after fitting each clip separately

DynEGA is included to test whether indicators driven by the same latent process can be identified through their coordinated changes over time (Boker et al., 2010; H. Golino et al., 2020; H. Golino et al., 2022; H. F. Golino & Epskamp, 2017). It estimates first derivatives from time-delay embeddings, constructs a regularized partial-correlation network among those derivatives, and identifies communities as candidate dimensions. GraphicalVAR (Epskamp et al., 2018; Epskamp, 2024) fits a multilevel vector autoregression that separates lagged from contemporaneous relations; its contemporaneous residual partial-correlation network is used for grouping.

GIMME (Gates & Molenaar, 2012; Lane et al., 2019) fits a structural vector autoregression to each clip separately. This allows the fitted temporal and contemporaneous relationships to differ across clips. Because the generating indicator assignments are shared across clips, we examine whether that common grouping remains detectable in the residual associations after these separate fits. Residual covariance matrices from successful fits are averaged and submitted to the common community-detection procedure. This adapts GIMME's usual use for recovering individual-specific directed relationships to the recovery of a shared indicator partition.

Static EGA estimates a regularized partial-correlation network from the contemporaneous correlation matrix of indicator levels, ignoring the observation order, and applies the same community-detection step. It serves two purposes. Statistically, it shows how much of the grouping is recoverable from contemporaneous levels

alone; if most of it is, a dynamic model cannot add much for this recovery target. Computationally, it requires far less time than the dynamic methods. Comparing static EGA with DynEGA and GraphicalVAR therefore shows whether modeling temporal structure improves grouping enough to justify its computational cost in these conditions.

Primary design and outcomes. The primary design uses $T = 100$ observations per clip and contains 21,600 datasets. It crosses the six scenario families, $m \in \{2, 3\}$ dimensions, $N \in \{50, 100\}$ clips, three loading levels (0.40, 0.60, 0.80), three residual standard deviations (0.125, 0.25, 0.50), and 100 replications, with $k = 4$ indicators per dimension. Every method receives the same dataset within a condition and replication. The empirical videos motivate an additional short-record condition. MAFW videos contain a median of 35 scored observations, and uninterrupted segments have a median length of 13.5 observations. We therefore also evaluate recovery with $T = 25$, using 21,600 corresponding datasets. Louvain at resolution 1.0 is the primary community-detection algorithm; Leiden and Walktrap were applied to the same fitted networks as sensitivity analyses.

The primary outcome is the overall correct-dimension recovery rate: the proportion of datasets in which the method returned a valid network and the estimated number of dimensions equaled m. A failed or timed-out fit counts as an unsuccessful recovery. We also report the valid-output rate and the correct-dimension rate among valid outputs, so that computational feasibility and accuracy can be examined separately. Assignment accuracy, the adjusted Rand index, and variation of information compare the full estimated partition of indicators with the true partition among valid outputs. Secondary diagnostics assess alignment between the recovered groups and the activity-scaled latent trajectories using best-match absolute trajectory correlation and trajectory NRMSE. Appendix B in the supplementary material defines how these diagnostics are constructed and matched.

Additional analyses. Three additional analyses examine the limits of the primary result. First, one-at-a-time sensitivity analyses start from a reference condition ($N = 100$, $T = 100$, loading 0.60, residual SD 0.25, common fixed initialization) and alter one design component at a time by changing the initialization rule, smoothing activity transitions, adding small cross-loadings, correlating residuals, correlating acceleration inputs, or transforming the indicators to bounded scores. The alternative initialization uses independent stationary starts for stable dimensions and bounded independent starts for amplifying dimensions. The bounded-score condition applies the softmax function directly to the p continuous indicators and analyzes either the p scores or their centered log-ratio (CLR) components. Because the transformed scores must sum to one, the original block structure among the p variables need not be preserved. Second, a small analysis adds clip-specific baseline and scale differences, with baselines constrained to the column space of the loading

matrix so that they add no structure outside the factor model. Third, an extended factorial simulation of 518,400 datasets evaluates DynEGA alone over additional numbers of dimensions, indicators per dimension, and observation schedules. Appendix B in the supplementary material gives the exact settings for every analysis.

Results

Study 1 Results: Similarity of Simulated and Real Score Series

Figure 4 compares the two generators with the MAFW reference. LOMM achieved a median plausibility of 0.970 (95% simulation range [0.935, 0.990]) and coverage of 0.920 [0.900, 0.950]. The corresponding MAFW-to-MAFW reference values were 0.950 [0.910, 0.990] for both measures. The static logistic-normal generator achieved plausibility of 0.510 [0.410, 0.590] and coverage of 0.370 [0.290, 0.455]. LOMM therefore generated score series that usually had a close MAFW counterpart and represented most MAFW evaluation series, whereas static calibration alone covered substantially less of the empirical variation.

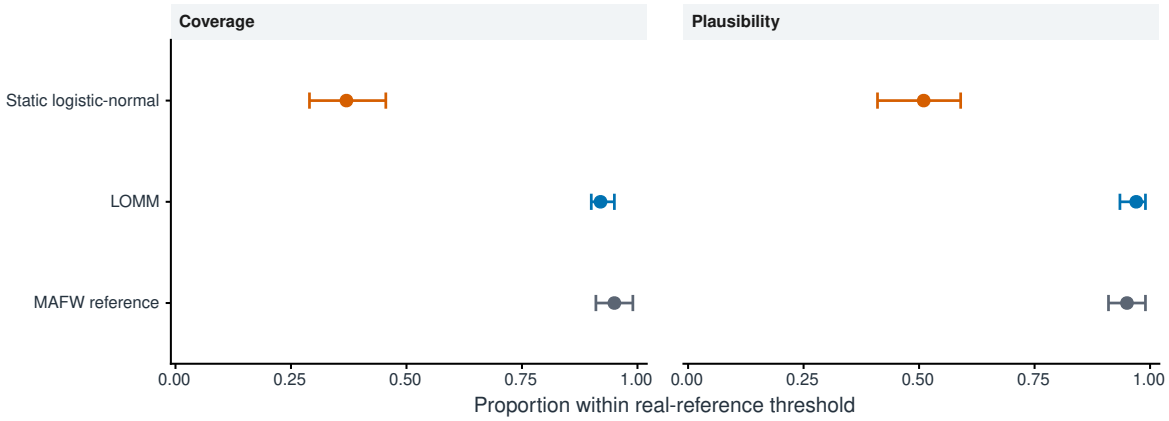


Figure 4: Cross-fitted similarity of LOMM and static logistic-normal series to MAFW evaluation series. Points are median proportions and bars are 2.5th and 97.5th percentiles across 100 generated datasets for each generator or 1,000 MAFW-to-MAFW reference comparisons. Plausibility is the proportion of simulated series with a nearest MAFW evaluation series within the fold-specific threshold. Coverage is the proportion of MAFW evaluation series with a nearest simulation within the same threshold. The threshold is the 95th percentile of directional nearest-neighbor distances between disjoint MAFW calibration samples. Higher values indicate closer agreement.

Across the metric-specific comparisons, 14 of the 22 LOMM feature discrepancies fell within the MAFW-to-MAFW reference range. Agreement was strongest for dispersion, short-term change, temporal persistence, and robust change variance. The clearest remaining differences concerned sample entropy, which was higher in LOMM series, and spectral entropy, which was lower. Appendix C in the supplementary material reports

all metric-specific results.

Study 2 Results: Recovery of Dimensionality and Indicator Assignments

Table 3 reports the $T = 100$ results under Louvain. The overall recovery rate requires both a valid network and the correct number of dimensions, so every failure and timeout counts against it. The last two columns report assignment accuracy among valid outputs and median runtime.

Table 3: Results for the primary $T = 100$ design under Louvain. Assignment accuracy is computed among valid network outputs. Runtime is the median, in seconds, among valid network outputs and includes model fitting, network extraction, and community detection.

Method	Valid output	Correct given valid	Overall recovery	Assignment accuracy	Median runtime, s
DynEGA	1.000	0.939	0.939	0.965	3.69
Static EGA	0.997	0.887	0.884	0.922	2.15
GraphicalVAR	0.941	0.826	0.777	0.902	85.41
GIMME	0.224	0.783	0.176	0.877	518.16

DynEGA returned a valid network for all 21,600 datasets and had the highest overall recovery rate, 0.939. Static EGA followed at 0.884, which shows that most of the grouping was recoverable from contemporaneous indicator levels. GraphicalVAR reached 0.777. GIMME recovered the correct dimension in 0.783 of its valid outputs but returned a valid network for only 0.224 of datasets, so its overall recovery was 0.176. Assignment accuracy among valid outputs was 0.965 for DynEGA, 0.922 for static EGA, 0.902 for GraphicalVAR, and 0.877 for GIMME. Static EGA was about 40 times faster than GraphicalVAR and had higher overall recovery and assignment accuracy for this target. GraphicalVAR estimates a more complex model that includes lagged effects; the comparison evaluates only whether that additional structure improved recovery of the grouping under these conditions. Appendix C in the supplementary material reports Monte Carlo standard errors, bootstrap intervals, and the remaining partition diagnostics.

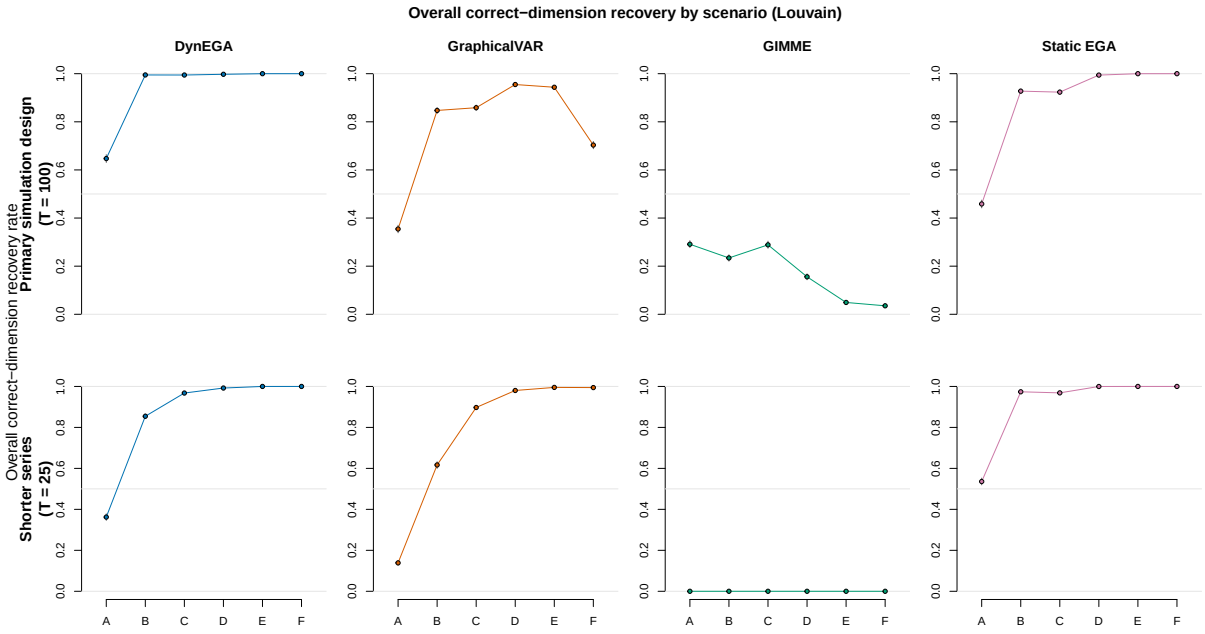


Figure 5: Overall correct-dimension recovery under Louvain by scenario family (codes as in Table 1), method, and observations per clip ($T = 100$, top; $T = 25$, bottom). A recovery requires a valid network output and the correct number of dimensions. Error bars are 95% bootstrap intervals from resampling whole datasets.

Recovery varied across scenario families and observations per clip (Figure 5). DynEGA had the highest recovery in most conditions. Static EGA's recovery was high because the block structure was present in contemporaneous levels. GraphicalVAR's recovery was lowest when expression was weak or measurement error was large. GIMME's low overall rates mainly reflected missing valid outputs. In the extended factorial simulation, DynEGA recovered the correct number of dimensions in 93.7% of 518,400 datasets and assigned 95.2% of indicators correctly; stronger loadings and smaller residual error improved recovery. Leiden and Walktrap led to the same conclusions as Louvain.

Secondary diagnostics assessed alignment between the recovered groups and the activity-scaled latent trajectories. Among valid outputs with the correct number of dimensions at $T = 100$, the best-match absolute trajectory correlation and trajectory NRMSE were 0.760 and 0.602 for DynEGA, 0.791 and 0.568 for GraphicalVAR, 0.606 and 0.898 for GIMME, and 0.787 and 0.563 for static EGA. GraphicalVAR and static EGA had higher trajectory correlations and lower NRMSE than DynEGA, while DynEGA had higher correlation and lower NRMSE than GIMME. These descriptive comparisons condition on the outputs satisfying each method's dimensional recovery criterion. Grouping recovery and trajectory recovery are different targets, and neither ordering ranks the methods in general. Appendix C in the supplementary material reports the

complete diagnostics.

Robustness and Sensitivity Analyses

Shorter records. With $T = 25$, static EGA had the highest overall recovery, 0.913, followed by DynEGA, 0.863, and GraphicalVAR, 0.770. Shortening the record slightly reduced DynEGA's recovery, left GraphicalVAR's about the same, and increased static EGA's. At $T = 25$, all 21,600 GIMME attempts ended in errors and none returned a valid network; none was recorded as a timeout. Among outputs with the correct number of dimensions, DynEGA's trajectory correlation was 0.848 and NRMSE was 0.521; GraphicalVAR had the highest correlation and lowest NRMSE, followed by DynEGA and static EGA (Appendix C in the supplementary material).

Initial states. In the reference condition of the one-at-a-time analyses ($N = 100$, $T = 100$, loading 0.60, residual SD 0.25, common fixed initialization), overall recovery was 0.993 for DynEGA, 0.912 for static EGA, and 0.683 for GraphicalVAR. Replacing the common fixed initialization with independent stationary starts for stable dimensions and bounded independent starts for amplifying dimensions produced the largest change among the sensitivity conditions: recovery fell to 0.838, 0.525, and 0.253, respectively. The common fixed initialization was therefore optimistic relative to this alternative.

Other one-at-a-time changes. Gradual activity transitions, small cross-loadings, and correlated acceleration inputs changed recovery little. A residual correlation of 0.10 between every pair of indicators increased recovery for GraphicalVAR and static EGA. This result shows that the particular residual pattern made the target grouping easier to recover; it does not establish robustness to residual dependence more generally.

Bounded scores. Transforming the p indicators to scores that sum to one changed the variables being analyzed and sharply reduced recovery of the original grouping. DynEGA recovered the correct number of dimensions in 0.333 of datasets analyzed as CLR components and 0.233 analyzed as raw scores; the corresponding rates were 0.333 and 0.167 for static EGA and 0.087 and 0.002 for GraphicalVAR. The softmax transformation and the accompanying projection onto the zero-sum subspace did not preserve the structure among the original indicators. For this reason, the Study 2 recovery results apply to continuous indicators and are not extended to bounded scores of the kind analyzed in Study 1.

Between-clip baseline and scale. This analysis used stable families A through D with independent stationary initial states. Its reference recovery rates are lower than those in analyses using the common fixed initialization. Adding clip-specific baseline and scale changed DynEGA's recovery from 0.775 to 0.750, a paired difference of -0.025 (95% interval $[-0.100, 0.050]$). Static EGA rose from 0.275 to 1.000: the added baselines lie in

the column space of the loadings, so they add between-clip variation that follows the intended grouping. GraphicalVAR and GIMME were unchanged, but those estimates rest on very few valid outputs; GraphicalVAR produced three correct outputs among 80 datasets, and every three-dimensional GIMME fit timed out.

Runtime and feasibility. Median runtime among valid $T = 100$ outputs was 2.15 s for static EGA, 3.69 s for DynEGA, 85.41 s for GraphicalVAR, and 518.16 s for GIMME. GraphicalVAR timed out on 1,271 of 21,600 fits. GIMME produced 4,846 valid outputs, 14,477 timeouts, and 2,277 post-estimation errors. DynEGA returned a valid output for every fit in the primary and extended designs. These runtimes are specific to our implementation and hardware. Appendix C in the supplementary material gives the full runtime, factorial, and community-detection results.

Discussion

LOMM separates three determinants of multivariate time-series structure: the dynamics of latent processes, their expression over time, and their measurement by observed indicators. This separation matters when researchers use associations among classifier-score channels to infer dimensional structure. A change in the recovered grouping may reflect a different latent structure, weaker expression of an unchanged structure, or properties of the measurement process. LOMM allows these possibilities to be examined under controlled conditions by holding the generating dimensions and indicator assignments fixed while varying dynamics, activity, or measurement quality.

The two studies establish different parts of this contribution. Study 1 supports the statistical resemblance of calibrated bounded output to short MAFW classifier-score series. Study 2 establishes recovery performance for specified continuous-indicator models. The initialization and bounded-score analyses show why those recovery results require explicit conditions: changes in the starting states or observation transformation can substantially change the recovered structure.

Study 1 found that LOMM generated classifier-score series with high plausibility and broad coverage relative to the MAFW reference. The static logistic-normal generator preserved the calibrated score geometry and between-video heterogeneity. It omitted temporal dependence and performed substantially worse on both measures. In this comparison, LOMM reproduced the joint statistical profile of the MAFW series more closely than the calibrated static logistic-normal generator. The metric-specific results also identify a concrete target for model development: LOMM captured most measures of level, dispersion, change, persistence, and cross-channel variation, while its entropy profile remained less similar to MAFW.

Study 2 evaluated recovery of a known factor structure from continuous indicators. DynEGA had the

highest overall recovery in the primary matched-method design and was feasible at the scale of the extended factorial simulation, where it was the only method evaluated. Static EGA showed that most of the grouping was recoverable from contemporaneous levels; it exceeded GraphicalVAR in both dimensional recovery and assignment accuracy while running in a median of 2.15 s rather than 85.41 s. For this grouping target and these conditions, GraphicalVAR's additional lagged model did not improve recovery relative to static EGA. GraphicalVAR answers additional questions about lagged relations, which this comparison does not evaluate.

The trajectory diagnostics in Appendix C in the supplementary material show that the ranking of methods depends on the recovery target. DynEGA's advantage concerned recovery of the number of dimensions and the indicator partition; alignment of recovered-group summaries with the latent trajectories among valid outputs with the correct number of dimensions produced a different ordering. The two targets should be evaluated separately. All comparisons apply to the specific estimation, network-conversion, and grouping procedures used here. GIMME's low overall recovery mainly reflected its failure to return valid outputs under a clip-level fitting procedure that differs from its usual use.

The most important qualification of the primary result concerns initialization. Recovery was lower with independent stationary starts for stable dimensions and bounded independent starts for amplifying dimensions than with the common fixed initialization. The primary initialization was therefore optimistic relative to this alternative. Because this sensitivity changed cross-dimension and cross-clip coherence together with the marginal initial-state distribution, it does not isolate the effect of sharing alone. Gradual activity transitions, small cross-loadings, and correlated acceleration inputs changed little. The bounded-score condition set a clearer limit: the softmax transformation changed the associations among the original indicators, and neither the raw scores nor the CLR components preserved the original grouping. The Study 2 recovery claims therefore apply to continuous indicators.

Limitations. Several limitations remain. The latent dimensions are uncoupled. The amplifying families produce large amplitudes late in the record and test recovery under those conditions only. The $T = 300$ conditions also change the observation interval, so they do not isolate the number of observations per clip. The GIMME residual-covariance conversion was constructed for this comparison and differs from typical idiographic use. Study 1 uses 100 short MAFW videos, one zero-shot scoring pipeline, and one LOMM specification chosen during model development on the same sample. LOMM is used here only as a generator with prespecified parameters; estimating it from data would require the identification constraints listed in Appendix A in the supplementary material.

LOMM provides a generative basis for studying how latent dynamics, changing expression, and measurement

jointly shape multivariate psychological time series. The present studies establish two initial uses: generating bounded series that reproduce selected statistical properties of video-derived emotion scores, and evaluating dimensional recovery from continuous indicators with a known generating structure. Together, they demonstrate how explicit assumptions about the generating processes can support controlled evaluations of psychological measurement. The bounded-score results also identify a central methodological challenge: transforming latent signals into proportions can change the observable structure on which dimensionality methods rely.

A next step is to develop procedures for fitting LOMM to observed data and recovering its dynamic, activity, and measurement parameters. State-space and hierarchical continuous-time methods provide starting points (Driver & Voelkle, 2018; Ou et al., 2019), with parameter-recovery studies needed to establish identification and estimation accuracy under realistic sampling and measurement conditions. A particularly relevant direction is the modeling of compositional time series, including gaze allocation across areas of interest (Fačevicová et al., 2026), daily allocations of time among sleep and activity behaviors (Le et al., 2022), and topic proportions in repeated language samples (Blei & Lafferty, 2006). In each case, increasing one component's share necessarily reduces the combined share of the others. LOMM's compositional observation layer offers a basis for simulating such dependent trajectories while controlling the latent processes that generate them. Extending this approach to estimation would allow researchers to investigate which features of psychological change remain identifiable when observations express relative allocations. This would develop LOMM into a broader framework for connecting theories of psychological dynamics with the constraints of compositional measurement.

Declarations

Funding

The authors received no specific funding for this work.

Competing interests

The authors declare that they have no competing interests.

Ethics approval

This study used simulations and secondary video data obtained under the MAFW research-use terms. It involved no new recruitment, intervention, or direct collection of human-participant data by the authors.

Consent to participate

Not applicable.

Consent for publication

The manuscript includes four still frames from MAFW solely to illustrate the score-extraction procedure. The frames are cited to the dataset, and no source video is redistributed. Their inclusion follows the dataset's terms for limited reproduction in academic publications.

Availability of data and materials

The project GitHub repository at https://github.com/atomashevic/LOMM contains the simulation summary results, analysis-ready tables underlying the figures, and aggregate MAFW-derived summaries used in the simulated-versus-real comparison. Raw MAFW videos and frame-level classifier scores are not redistributed. Access to MAFW is governed by the dataset provider, and an end-to-end reanalysis of Study 1 requires approved access and local generation of the classifier scores.

Code availability

Code for generating LOMM datasets, fitting the methods, reproducing the statistical summaries, and regenerating the tables and figures is available in the same GitHub repository. The repository also provides the manuscript source and instructions for reproducing the analyses.

Authors' contributions

Aleksandar Tomašević contributed conceptualization, formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Hudson Golino contributed formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Alexander P. Christensen contributed methodology, software, writing the original draft, and reviewing and editing the manuscript.

Open Practices Statement

The study was not preregistered. The simulation and analysis code, manuscript source, paper-facing tables, and aggregate MAFW-derived summaries used in the simulated-versus-real comparison are available at https://github.com/atomashevic/LOMM. Raw MAFW videos are third-party materials, and the frame-level classifier scores are derived from those restricted materials; neither is included.

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